GeoFusionNet / GeoCorr: Revised Research Direction

Proposal for approval · 2026-10-10 · main @ 5572daf · Gobika S

Summary. The original thesis was that a monocular-anchored geometry prior, fed to the network as an input (GeoFusionNet's GGF) and then used to correct the matching score (GeoCorr Lite), would fix the failures of self-supervised MVS. The first half was tested and gave no measurable gain. The prior is nearer the true cost dip on only **52.4%** of failure pixels and within 4 mm of ground truth on only **11.5%**. The frozen Stage 0 baseline A0f scores **0.437 / 0.545 / 0.491** (acc / comp / overall, mm, 3 seeds × 22 scans, 1600×1152). The one large measured gain came from capacity: C1_capacity scores 0.418 (seed 2) and 0.414 (seed 0), against 0.479 for A0f seed 2.
The revised direction makes the **empirical study of what limits self-supervised MVS** the primary track (Track 2). The frozen-monocular-feature run is an exploratory screen (Track 1). **GeoCorr Lite (Stage 2) has never been run.** It is proposed as the next experiment, not reported as a result.

Conventions. All numbers are measured, from `runs/`, unless marked **PENDING** (not yet measured) or **NOT RUN**. Fused point-cloud scores use DTU test (22 scans), 1600×1152 input, GeoMVSNet's fusion and the official DTUeval protocol, in mm (lower is better), unless stated otherwise. **Naming:** the queue's run `B2_capacity` is reported here as **C1_capacity** (C1). This avoids a collision with the manuscript's B2, which is GeoCorr Lite. Run directories keep their original names (`runs/B2_capacity/`, `runs/B2_nogeo/`, `runs/B2_mono_feat/`). B2_nogeo and B2_mono_feat are C1-size models.

1. What was done, from the start

1.1 The original plan (manuscript thesis)

Thesis: geometry-aware self-supervised MVS. A self-supervised cascade network (GeoFusionNet) is given an external metric geometry prior G. G enters in two ways, each added as a zero-initialized residual so that any gain can be attributed:

- **GeoFusionNet (Stage 1):** geometry-guided feature fusion (GGF). A separate geometry encoder turns G into features that are added to the reference-view features *before* the cost volume (representation level). L_geo, a loss pulling predicted depth toward G, is also part of the objective.
- **GeoCorr Lite (Stage 2):** a residual on the raw matching-score volume, `score + α · gate · energy`. The energy comes from a few scalar geometric compatibility maps through one shared 1×1 head, and α starts at 0. This changes *which depth hypothesis wins* (scoring level).
- **Staged rule:** no architectural comparison counts until Stage 0 is frozen and measured over 3 seeds.

Prior G: Depth-Anything-V2-Small relative inverse depth, fitted per view to metric mm by RANSAC on `1/z = s·g + t` against SIFT points triangulated with the known DTU cameras (no ground truth is used). Views with too few anchors, mainly on textureless scans, get no prior.

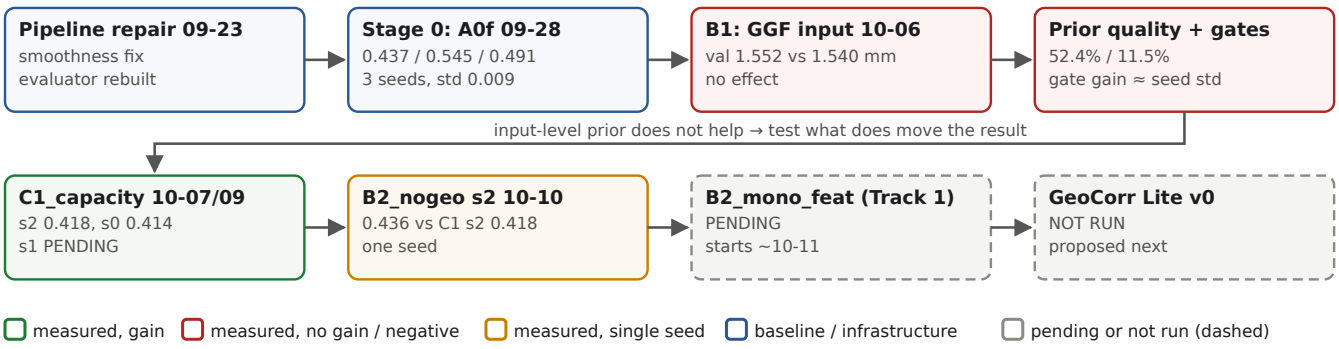


Figure 1. Project history as a flow. Every box is a measured result except the dashed ones (PENDING or NOT RUN). Scores are acc / comp / overall in mm under our evaluator (§5).

1.2 Timeline of measured results

Date	Step	Measured result	Verdict
09-23	Pipeline repair	Constant-depth failure (~67 mm) traced to smoothness on raw mm depth and fixed by mean-normalizing. Evaluator rebuilt to GeoMVSNet fusion + official DTUeval. Earlier numbers are void.	Fixed
09-24/25	A0 baseline, 3 seeds	0.447 / 0.586 / 0.517 (1600×1152). At the 640×512 train crop: 0.379 / 0.732 / 0.556.	Superseded
09-25	Evaluator comparison: CL-MVSNet point clouds under the official metric	Point clouds generated locally at 640×480 (not released point clouds): 0.446 / 0.431 / 0.438 vs 0.329 published. The 0.438 vs 0.329 gap mixes test resolution (640×480 here vs the 1184×1600 set in CL-MVSNet's test script), the evaluation crop, fusion thresholds and checkpoint differences; its cause is unconfirmed.	Evaluator sensitivity
09-25	CL-MVSNet LR recipe fine-tune (6k steps)	Val fine median 1.75 → 1.65–1.70 mm, all by step 1000.	~5%; LR not the cause
09-25	Cost-volume fixes (out-of-view mask, trilinear up-sampling)	Sawtooth probabilities 41% → 0.1% of wrong pixels.	Adopted
09-25	Cost-profile probe, scan48	GT is the global cost minimum at 30% of wrong pixels (89% of right ones). Our photometric loss prefers GT at only 42% of wrong pixels; CL-MVSNet's L0.5 min-view loss at 12%.	Photometric ambiguity
09-27	CL-MVSNet ICC+SCC consistency (A0c) vs control (A0f)	Aggregate val within 0.05 mm (noise); scan48 fine median 4.34 vs 3.45 mm (worse).	Dropped
09-27/28	**Stage 0 frozen: A0f**, 3 seeds	**0.437 ± 0.003 / 0.545 ± 0.016 / 0.491 ± 0.009**	Baseline
10-06	B1: G as input via zero-init GGF (matched control)	Val fine 1.552 vs 1.540 mm; scan48 competing-dips share 18.53 vs 18.42% of valid pixels.	**No effect**
10-06	L_geo off at A0f size (matched control)	Val fine 1.565 vs 1.540 mm; dips 18.57 vs 18.42%.	Noise
10-06	Prior quality, 22 test scans	Median |G−GT| 5–69 mm per scan (model 0.8–4 mm). On ambiguous pixels G is nearer the true dip on 52.4%, within 4 mm on 11.5%.	Coin flip
10-06	Prior confidence gates, val-frozen	Only reversed coarse disagreement (|G − coarse| ≥ 99.3 mm) passes: test 21.2% coverage, 86.7% correct.	One gate passes
10-07	Gate through fusion, 22 scans	A0f s2 0.479 → 0.470; C1 s2 0.418 → 0.415.	Gain ≈ seed noise
10-07	**C1_capacity**, seed 2	0.473 / 0.362 / 0.418, better than A0f s2 on 17/22 scans; per-view depth not better.	**Large fused gain**
10-07	Fusion pass-through check	Confidence filter ≈ no-op. At A0f's pass rate C1 still scores 0.416 (fewer points).	Gain not from point count
10-09	C1_capacity, seed 0	0.470 / 0.358 / 0.414	Replicates
10-10	B2_nogeo, seed 2 (C1 size, L_geo off)	0.461 / 0.410 / 0.436 (vs C1 s2 0.418)	See §4.1(c)
10-09	Monocular-feature injection (B2_mono_feat) implemented	Patch fc8c805: frozen DA-V2-S reference features, zero-init projection; identity at step 0. Queued only.	Not yet measured

10-10	C1_capacity, seed 1	**PENDING**: running; base phase at step 5,050 / 20,000 at 10:47 UTC (started 07:50 UTC); queue ETA ~10-11 07:30 UTC
—	B2_mono_feat, seed 2 (Track 1)	**PENDING**: queued behind the C1 queue, starts ~10-11 07:30 UTC
—	GeoCorr Lite (Stage 2)	**NOT RUN**: the code exists but no experiment has used it

1.3 Headline fused scores

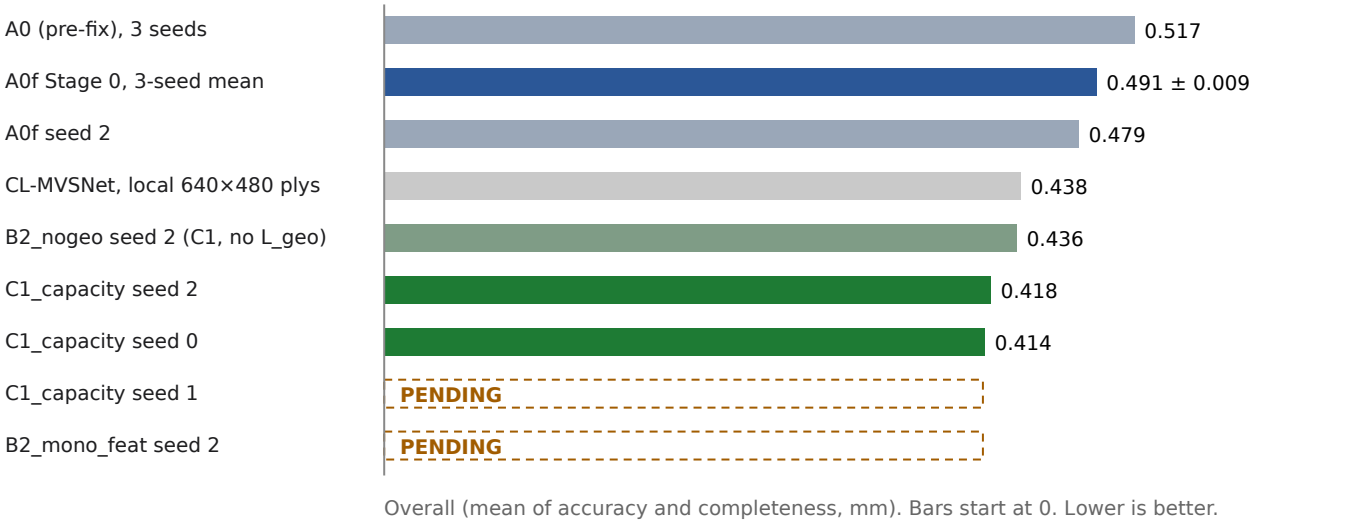


Figure 2. Headline fused overall scores (mm, lower is better) under our evaluator, from the run JSONs. Dashed bars are PENDING.

Model	Params*	Acc	Comp	Overall	Notes
A0 (pre-fix), 3 seeds	215k	0.447	0.586	0.517	Superseded
A0f Stage 0, 3 seeds	215k	0.437 ± 0.003	0.545 ± 0.016	**0.491 ± 0.009**	Population std over seeds
A0f seed 0 / 1 / 2		0.436 / 0.441 / 0.435	0.558 / 0.555 / 0.523	0.497 / 0.498 / 0.479	
C1_capacity seed 0	2.12M	0.470	0.358	0.414	
C1_capacity seed 2	2.12M	0.473	0.362	0.418	Better than A0f s2 on 17/22 scans
C1_capacity seed 1	2.12M			**PENDING**	Running
C1_capacity 3-seed mean ± std				**PENDING**	After the queue finishes
B2_nogeo seed 2 (C1, L_geo off)	2.12M	0.461	0.410	0.436	Matched to C1 s2
B2_mono_feat seed 2 (Track 1)				**PENDING**	Matched control: B2_nogeo s2
CL-MVSNet point clouds generated locally at 640×480, our metric (§5.3)		0.446	0.431	0.438	Published 0.329; not reproduced

* Parameter count as reported by the evaluator (`param_count`), which includes the geometry encoder. In A0f and C1 that encoder runs but its output is discarded (GGF off). Parameters on the depth path: about 121k (A0f) and 2.03M (C1).

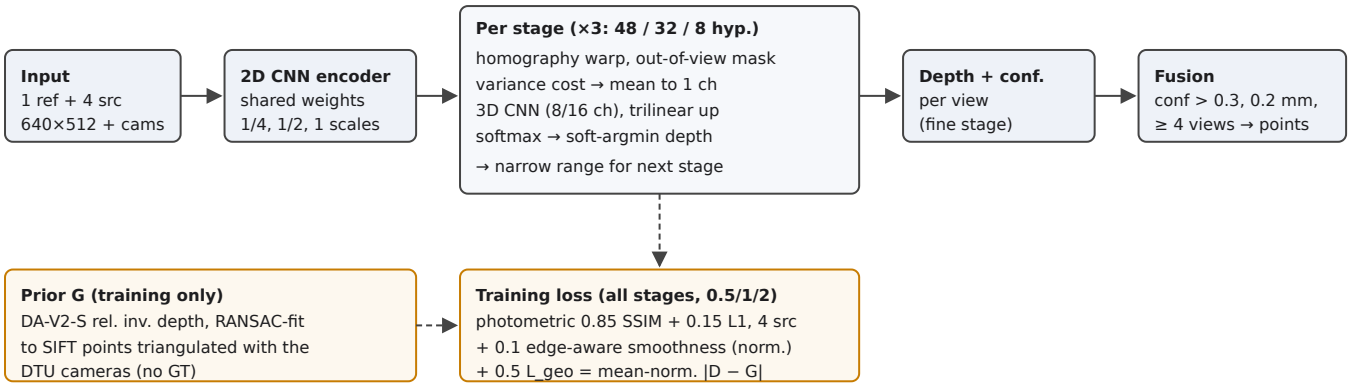
2. The baseline and the current model

2.1 Architecture

Both models are 3-stage CasMVSNet-style cascades over 1 reference + 4 source views at 640×512 in training. Each stage warps source features onto fronto-parallel depth planes with a differentiable homography, builds a cost volume, regularizes it with a 3D CNN, and regresses depth by soft-argmin (softmax expectation). Later stages search a narrowed range around the up-sampled previous depth.

Component	A0f (Stage 0, frozen)	C1_capacity (current model)
Feature encoder	Small 2D CNN, 3 scales, base 32 channels	FPN, widths 32/96/192, inner 64
Cascade	coarse 1/4 res, 48 hypotheses over the full range (interval ×4) → mid 1/2 res, 32 hypotheses (×2) → fine full res, 8 hypotheses (×1); base interval 2.65 mm	
Warping	Homography warp; out-of-view source samples masked (fix adopted 09-25)	
Cost volume	Variance over 5 views, averaged to 1 channel	Group-wise variance, 8 groups
3D regularizer	2 levels, 8/16 channels, trilinear up-sampling	3D U-Net, base 16, 2 levels, trilinear up-sampling
Depth	Soft-argmin per stage; confidence from the probability volume	
Prior G at train time	Used only by L_geo; the geometry encoder output is discarded (GGF off)	
Parameters (depth path / reported)	~121k / 215,211	~2.03M / 2,121,459

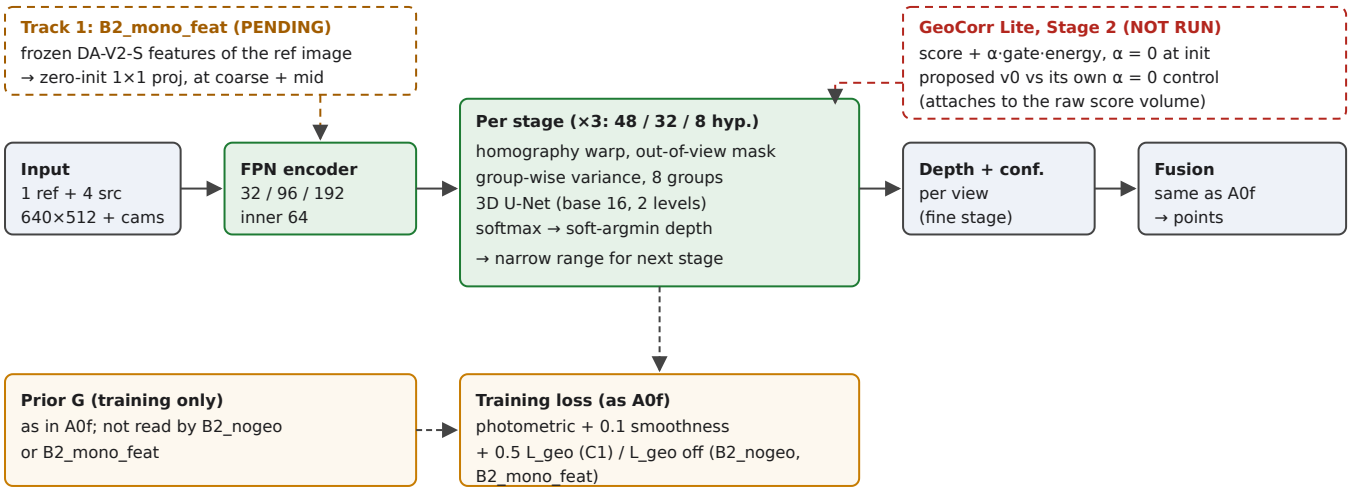
2.2 Block diagram: A0f baseline



Solid arrows: inference path. Dashed: training only. No ground-truth depth is used in training.

Figure 3. Block diagram of the A0f baseline (Stage 0, frozen).

2.3 Block diagram: current model (C1_capacity) and where the open work attaches



Green: what changed from A0f (C1_capacity). Orange dashed: Track 1 add-on, result pending.
Red dashed: GeoCorr Lite, never run.

Figure 4. Block diagram of the current model (C1_capacity) and where Track 1 and GeoCorr Lite would attach.

2.4 Training recipe

Phase	A0f chain (config)	C1 chain (config)	Settings
1. Base	A0_baseline	B2_capacity_base	20k steps, Adam, constant lr 1e-3, batch 2
2. Fine-tune	A0_finetune_clmvs	B2_capacity_finetune	6k optimizer steps, batch 2 × 2 accumulation, lr 5e-4 (CL-MVSNet recipe), 200-step warm-up, halved at 60/75/90%
3. Final	A0f_fixes_only_short	B2_capacity	1.5k steps, lr 5e-4, 100-step warm-up; cost-volume fixes on

- Data: DTU training split, 5 views, 640×512, all 7 lightings, 192 × 2.65 mm depth range. 18-scan validation split (882 views) is evaluated every 500 steps.
- Self-supervised: no ground-truth depth enters any loss. G is an auxiliary pretrained-model prior and is disclosed as such.
- In A0f the cost-volume fixes come on only in phase 3. In C1 they are on from step 0.
- For scale: A0 saw ~40k samples, about 9% of the reference recipes' 16 × 27,097.

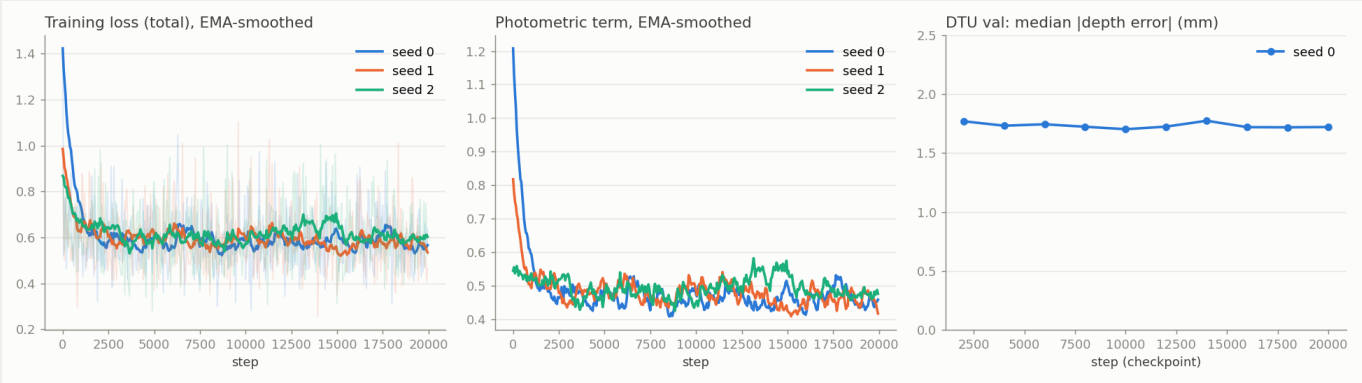


Figure 5. Training curves of the pre-fix A0 base phase (runs/A0_baseline/convergence.png, 3 seeds, 20k steps, constant lr 1e-3). Loss and DTU-val median depth error (~1.72 mm) are flat from ~2k steps. This plateau motivated the CL-MVSNet-recipe fine-tune (phase 2), which gave only ~5%.

2.5 Evaluation protocol

Item	Setting
Test set / input	All 22 DTU test scans; 1600×1200 resized to 1600×1152 (as GeoMVSNet's test script)
Fusion	Port of GeoMVSNet's DTU geometric-consistency fusion: confidence > 0.3, world-space point distance < 0.2 mm, ≥ 4 consistent views counting the reference. Fixed for all scans. Full details from the code: §5.
Metric	Official DTUeval-python (accuracy, completeness, overall = mean), cross-checked on scan1 against the unmodified reference script (§5)
Seeds	3 seeds for headline numbers (mean ± population std); single-seed comparisons use matched pairs (same init and sample order, one change)
Depth diagnostics	`costvol_diag.py`: per-stage median error, % > 4 mm, GT-in-window coverage. `costvol_profile_probe.py`: cost-curve classes at wrong pixels (fine error > 4 mm)
Validity	Only numbers produced on or after 2026-09-23 count

3. Why the direction changed

> **The geometry prior, fed through a separate encoder (GGF), did not help.** In a matched pair against an identical control, B1 changed validation and scan48 depth by less than the control's own checkpoint-to-checkpoint swing, and it did not move the competing-dips share. The measured reason: at the pixels where the model fails, the prior is not reliable enough to choose between the competing depth hypotheses.

3.1 What the failures are

The errors concentrate on scans 11, 13, 48, 62 and 77. On scan48 (A0f seed 2, 640×512), 42.9% of valid pixels are wrong (> 4 mm). At those pixels the raw cost curve has the GT depth as a *secondary* local minimum ("competing dips") on 18.75% of valid pixels, and the photometric loss itself prefers the wrong depth on most wrong pixels. Across every training change at A0f size, the competing-dips share stayed at 18.4–20.0%. The image evidence alone cannot decide, so the original plan turned to an external prior.

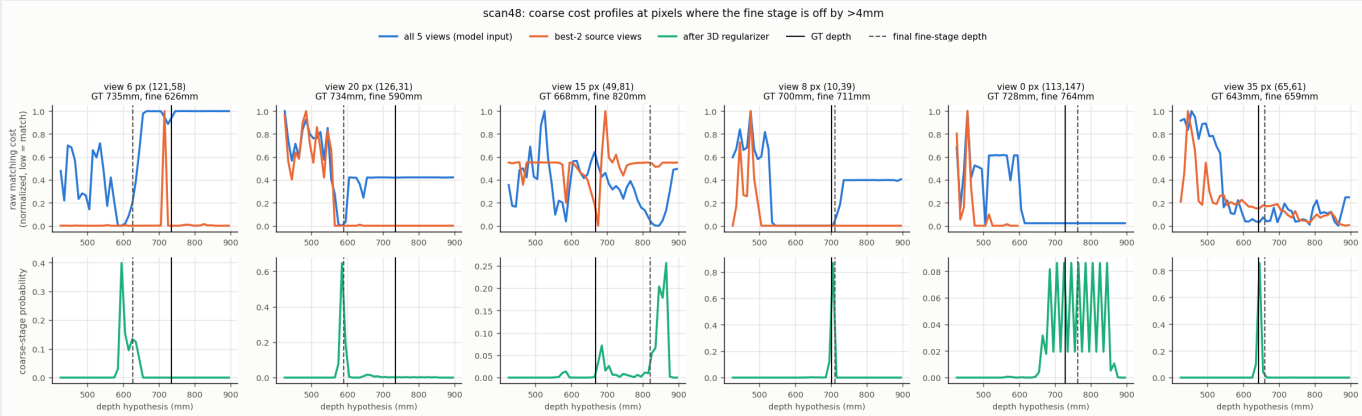


Figure 6. Raw cost profiles at scan48 pixels where the fine stage is off by > 4 mm (runs/A0_finetune_clmvs/seed2/probe/scan48_coarse_profiles.png; the A0 fine-tune checkpoint, the predecessor of A0f). Solid black = GT depth, dashed = predicted. The GT is often a secondary dip ("competing dips") or not a minimum at all, and the 3D regularizer commits to the wrong mode. Image evidence alone cannot decide these pixels.

3.2 Measured evidence against the prior as an input

Measurement (A0f seed 2)	Value	Source
Prior nearer the true dip than the chosen dip, failure (ambiguous) pixels, pooled over 22 scans	**52.4%** (21–72% by scan)	prior_vs_dips.py
Prior within 4 mm of GT, failure (ambiguous) pixels	**11.5%**	prior_vs_dips.py
Median dip separation / median prior error, ambiguous pixels	60 mm / 27.7 mm	prior_vs_dips.py
Prior closer to GT than the prediction, wrong pixels	61.5%	prior_vs_dips.py
Median |G − GT| per scan (model: 0.8–4 mm)	5–69 mm	prior_quality.py
scan48 views with no valid prior	16 of 49	prior_coverage.py
B1 (GGF input) − control: val fine median / scan48 fine median	+0.012 / −0.44 mm	control's own swing 0.39 mm
B1 − control: scan48 competing dips (% valid)	18.53 vs 18.42	probe
L_geo off − control (A0f size): val fine / scan48 fine	+0.025 / −0.20 mm	noise

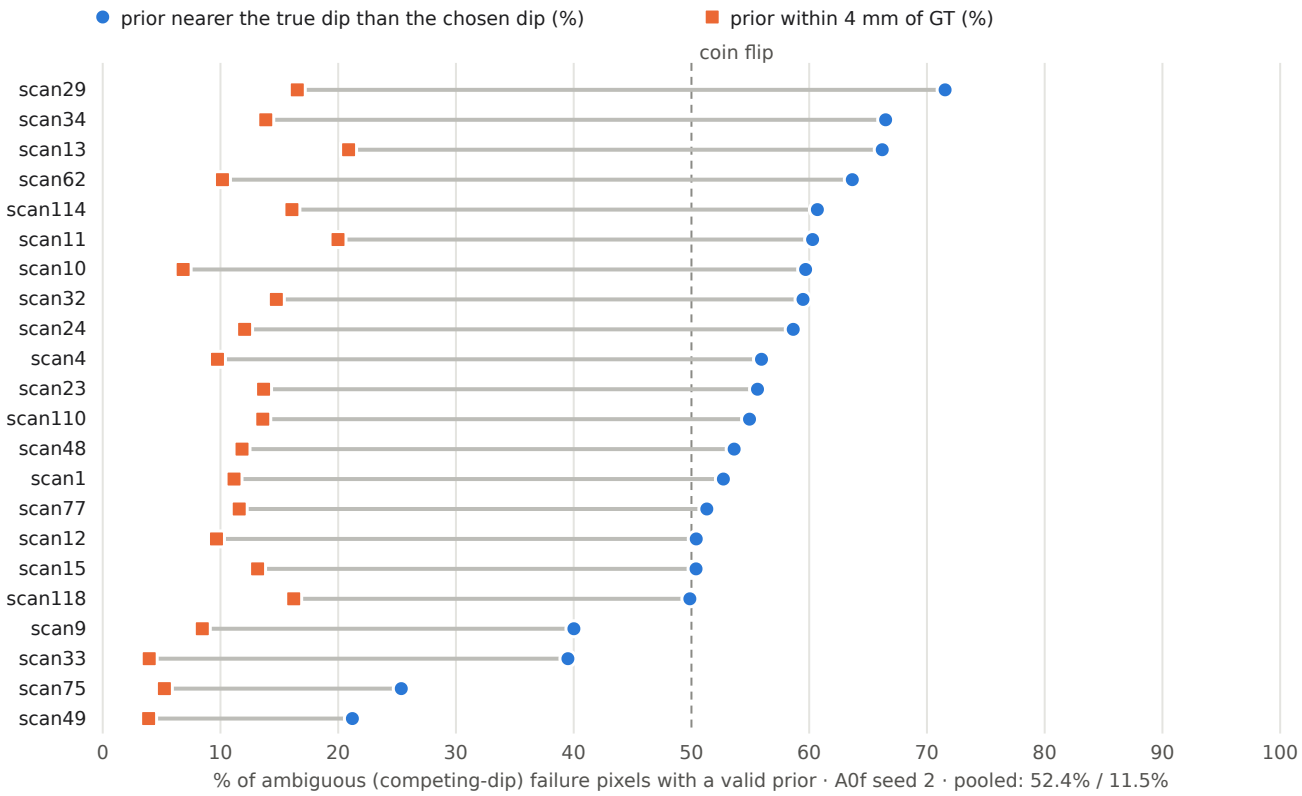


Figure 7. Per-scan prior quality at the failure pixels that matter (ambiguous pixels: the cost curve has competing dips), A0f seed 2, from runs/prior_vs_dips/A0f_seed2.json. Blue = how often the prior is nearer the true dip than the chosen one; orange = how often it is within 4 mm of GT. Pooled over 22 scans: 52.4% and 11.5%. Most scans sit near the 50% coin-flip line.

3.3 Gating the prior does not rescue it end to end

A gate was accepted only if, with thresholds fit on the 18 validation scans and frozen, it covered ≥ 20% of ambiguous test pixels at ≥ 80% correct-dip rate (pre-registered). Of six single signals (both orientations) and a logistic combination, only one passed:

Gate	Val cov / correct	Test cov / correct	Fused overall, ungated → gated
Reversed coarse disagreement, |G − coarse| ≥ 99.3 mm (A0f s2)	20.4% / 90.7%	21.2% / 86.7%	0.479 → 0.470
Same gate refit on C1 (≥ 89.3 mm, C1 s2)	20.0% / 86.3%	23.6% / 82.2%	0.418 → 0.415
Combined logistic (A0f s2)	20.4% / 95.3%	17.5% / 92.8% (fails)	—
All other signals	—	≈ 50–60% correct	—

The passing gate is partly mechanical: it fires where G disagrees with the model by ~100 mm, i.e. on gross outliers. The A0f gain (−0.009) equals A0f's seed std (0.009), and almost all of it comes from scan48 (−0.116) and scan77 (−0.114); the other 20 scans are slightly worse in sum (+0.024). On C1 the gain is −0.003.

3.4 What did move the result

Capacity. C1_capacity (17× more parameters on the depth path) improved fused overall from 0.479 to 0.418 at seed 2 and reached 0.414 at seed 0. This came with lower completeness error (0.362 vs 0.523). Per-view depth maps did not improve (§4.1(b)), and why C1's depths fuse better is not yet explained.

4. The new direction

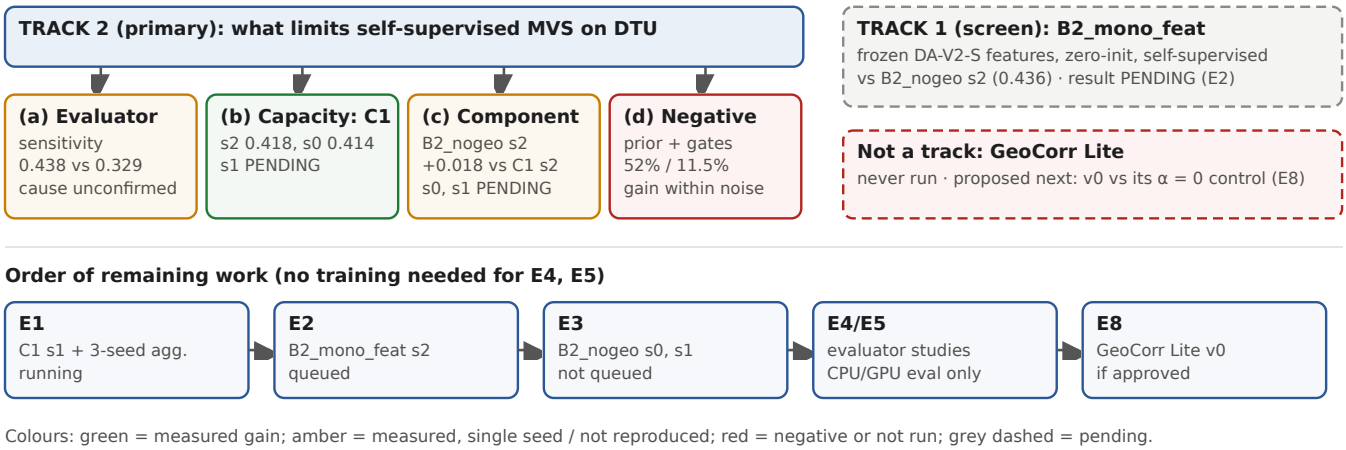


Figure 8. Structure of the revised direction: Track 2 (primary, four findings), Track 1 (screen), GeoCorr Lite (not a track, never run), and the order of the remaining experiments (§6.1).

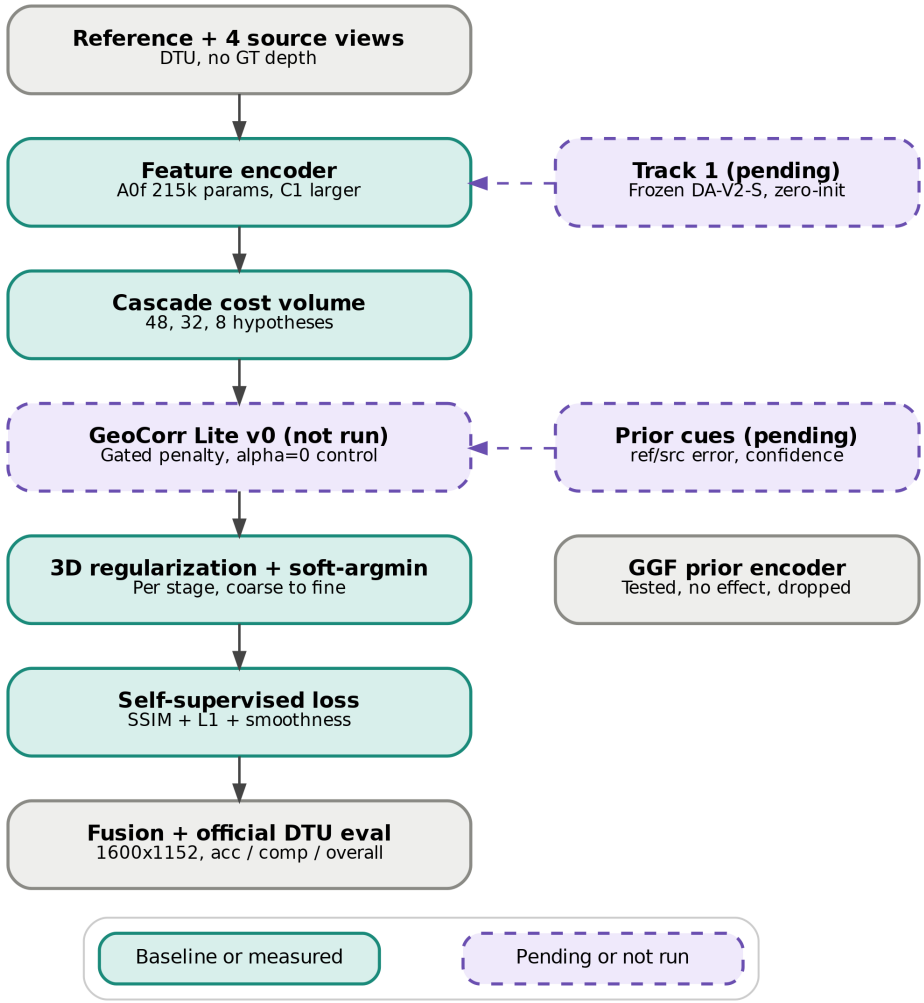


Figure 9. Proposed pipeline with the open work attached (reports/figs/proposal_block_diagram.png, graphviz, 300 dpi; source .dot and .svg alongside). Teal = baseline or measured; purple dashed = pending or not run. The GGF prior encoder is shown without arrows because it was tested and dropped.

Status of every item

Item	What it is	Status	Evidence (run files and logs, read 2026-10-10 10:53 UTC)
A0f baseline	Stage 0, frozen: small encoder, 3-stage cascade, 3-phase training	**measured**	Measured: 3 seeds (runs/A0f_fixes_only_short/aggregated_1600x1152.json, n = 3)
C1 capacity	A0f with FPN encoder, group-wise cost volume, larger 3D U-Net (queue name B2_capacity)	**measured** / **pending**	Measured: seeds 0 and 2 (eval_1600x1152.json). Pending: seed 1 running, base phase step 5,250 / 20,000 at 10:53 UTC (runs/queue_b2_seeds_nogeo.log)
B2_nogeo	C1 with L_geo weight 0 (matched to C1 seed 2)	**measured** / **pending**	Measured: seed 2 (runs/B2_nogeo/seed2/eval_1600x1152.json). Pending: seeds 0 and 1 not queued
GGF	Prior G through a zero-init geometry encoder into the reference features (B1)	**measured**	Measured at depth level only, seed 2 (runs/B1_ggf_input/seed2: val_depth.jsonl, costvol_diag, probe). No fused eval was run; dropped
Track 1 mono feat	Frozen DA-V2-S reference features, zero-init, self-supervised (B2_mono_feat)	**pending**	Pending: queue PID 3457546 waiting for the C1 queue; runs/queue_b2_mono_feat.log is empty; no runs/B2_mono_feat directory
GeoCorr Lite v0	Score-level gated penalty vs its alpha = 0 control	**not run**	Not run: configs exist (A3_geocorr_fixed, A4_learned_gate) but no run directory exists for any GeoCorr-enabled config
3-seed aggregate	Mean ± std over seeds 0, 1, 2	**measured** / **pending**	Measured for A0f (0.491 ± 0.009). Pending for C1 (no aggregate file; needs seed 1)

4.1 Track 2 (primary, ongoing)

TRACK 2 (primary, ongoing): a measured empirical study of what limits self-supervised MVS on DTU. Contributions are findings, not a new architecture:

(a) Evaluator sensitivity. CL-MVSNet point clouds generated locally at 640×480 from CL-MVSNet's released checkpoint (not released point clouds) score 0.438 under our evaluator vs 0.329 as published. The 0.438 vs 0.329 gap mixes test resolution (640×480 here vs the 1184×1600 set in CL-MVSNet's test script), the evaluation crop, fusion thresholds and checkpoint differences; its cause is unconfirmed. Separately, scoring at the 640×512 crop (~68% of the image area) instead of the full frame inflates completeness error (A0: 0.732 vs 0.586). Report both evaluators.

Evidence	Value	Status
CL-MVSNet point clouds generated locally at 640×480 (released checkpoint, its own filter; not released point clouds), official metric = our metric	0.446 / 0.431 / **0.438**	Measured
CL-MVSNet, published (its own fusion: conf 0.8, 3 source views, 0.001 relative depth, per-scan pixel threshold)	0.329	Published; not reproduced
Which setting accounts for the 0.109 gap (fusion settings vs test resolution, see §5.3)	—	**PENDING**: not yet isolated by an ablation
640×512 training crop vs full frame: 640×512 of the 800×600 MVSNet layout = 68.3% of the image area	—	Geometry of the crop
A0, 3 seeds: completeness at 640×512 crop vs 1600×1152	0.732 vs 0.586 (overall 0.556 vs 0.517)	Measured
A0f / C1 scored under the CL-MVSNet fusion protocol ("both evaluators")	—	**PENDING**

(b) Capacity (C1_capacity; queue name B2_capacity). Hypothesis under test: the ~215k-parameter A0f backbone limits the fused result. C1 scored 0.418 on seed 2 vs 0.479 for A0f. 3-seed results are PENDING until the queue finishes; report seed-level numbers only.

Seed-level (fused)	Acc	Comp	Overall	Scans better than A0f, same seed
A0f seed 2	0.435	0.523	0.479	—
C1 seed 2	0.473	0.362	0.418	17 / 22
C1 seed 0 (A0f seed 0: 0.497)	0.470	0.358	0.414	—
C1 seed 1			**PENDING**	
C1 3-seed mean ± std			**PENDING**	

Per-view depth (640×512, final checkpoints). C1 lowers the competing-dips share but its depth maps are not more accurate:

Metric	A0f s2	C1 s0	C1 s2	B2_nogeo s2
Val fine median error (mm, 882 views)	1.52	1.71	1.76	1.81
Val fine GT coverage (%)	76.8	77.2	76.9	74.0
scan48 fine median error (mm)	3.45	4.13	3.96	4.57
scan48 pixels > 4 mm (%)	47.5	50.7	49.8	52.5
scan48 fine GT coverage (%)	62.6	65.0	66.7	61.6
scan48 competing dips (% valid)	18.75	13.59	12.90	14.96
scan48 no minimum at GT (% valid)	11.17	12.74	12.64	13.04

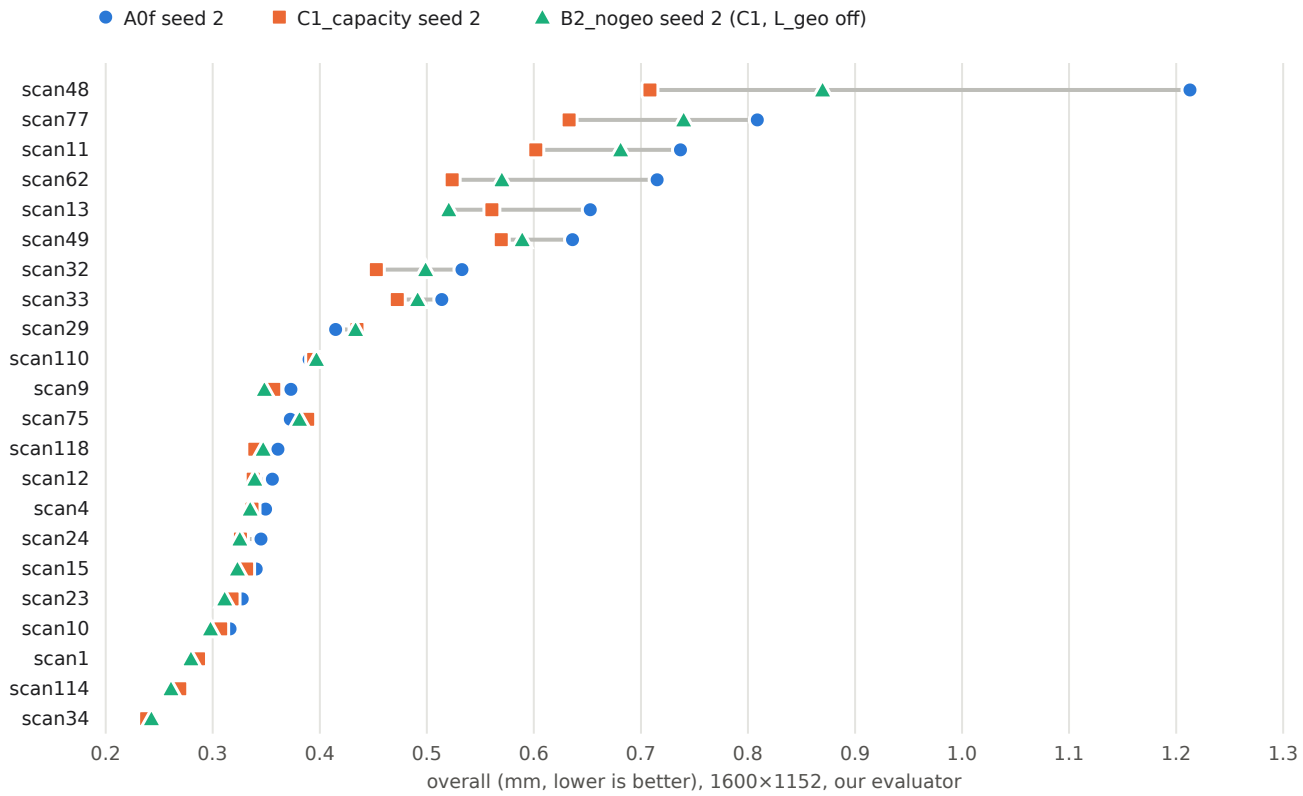


Figure 10. Per-scan fused overall (mm) for the three finished seed-2 models, from each run's eval_1600x1152.json. Scans are sorted by A0f seed 2 (worst first); the grey bar spans the three models. Capacity (orange square vs blue circle) helps most on the worst scans. Removing L_geo at C1 size (aqua triangle) loses part of that gain on scans 48, 77, 11. Seed-level only: C1 seed 1 and B2_nogeo seeds 0 and 1 are PENDING.

(c) Component analysis: B2_nogeo (prior loss off at C1 size), results from the finished runs, with seed noise (A0f std 0.009).

Seed 2, matched pair (only the L_geo weight differs: 0.5 vs 0)	Acc	Comp	Overall
C1_capacity (L_geo on)	0.473	0.362	0.418
B2_nogeo (L_geo off)	0.461	0.410	0.436
Difference (no-geo − C1)	−0.012	+0.048	**+0.018**

- Seed noise: A0f std 0.009 (3 seeds). The two finished C1 seeds differ by 0.004. The +0.018 is about 2× A0f's seed std, but it is **one seed**.
- B2_nogeo is better on 11/22 scans. The loss is concentrated on the outlier scans: scan48 0.708 → 0.870, scan77 0.633 → 0.740, scan11 0.602 → 0.681, scan62 0.524 → 0.570, scan32 0.453 → 0.499. scan13 goes the other way (0.561 → 0.521).
- Its depth maps are also worse than C1 s2 (val fine 1.81 vs 1.76 mm; scan48 4.57 vs 3.96 mm).
- At A0f size the same switch was noise (§3.2). B2_nogeo seeds 0 and 1: **PENDING** (not queued).

(d) Negative result: the monocular-anchored geometry prior (Depth Anything V2 Small, RANSAC-anchored to SIFT points) does not help. It is nearer the true dip 52% of the time and within 4 mm 11.5% of the time on failure pixels. Validation-frozen gates passed only for the reversed coarse-disagreement gate, and the end-to-end gain is within noise.

Evidence: §3.2–3.3 (B1 no effect; 52.4% / 11.5%; one gate passes; fused gain −0.009 on A0f, equal to the seed std, and −0.003 on C1).

Tension with (c), to resolve before the paper commits to (d) as stated. (d) holds for the prior as an *input* (B1/GGF) and as a gated *replacement*. The B2_nogeo pair, however, suggests the same prior used as a *training loss* (L_geo) may help at C1 size (+0.018 overall when removed, concentrated on the outlier scans). If B2_nogeo seeds 0 and 1 confirm this, (d) should be scoped to "as an input or test-time replacement". If they do not, (d) stands as written.

4.2 Track 1 (screening, exploratory)

TRACK 1 (screening, exploratory): frozen Depth-Anything-V2-S features injected as a zero-init encoder input in a self-supervised setting (no GT depth), the B2_mono_feat run. The result is PENDING. The closest work (MonoMVSNet, M2Depth) is supervised, so the narrow claim is only "does this help without GT supervision", and it is a claim only if the run shows a gain beyond seed noise. State that it is not yet established.

Item	Setting / status
Change vs control	Frozen DA-V2-S features of the reference image (518-px input, fp16), zero-initialized projection, added to the encoder features at the coarse and mid stages. Identical to the control at step 0 (identity test). No prior depth input, no L_geo.
Matched control	B2_nogeo seed 2 (same phase-2 checkpoint, same phase-3 recipe): overall 0.436
Result	**PENDING**: starts ~2026-10-11 07:30 UTC
Claim criterion	Gain beyond seed noise (A0f std 0.009). A single-seed gain would need more seeds before it counts.
Status of the claim	**Not yet established.**

4.3 Not a track: GeoCorr Lite

GeoCorr Lite (the manuscript's B2, Stage 2) has never been run. No number in this report or in any earlier report comes from it. It is proposed as the next experiment: a minimal v0 (scalar geometric compatibility maps → one shared 1×1 head → `score + α · gate · energy`, zero-initialized) compared against its own α = 0 control, matched seed and sample order. It is not a result and not a track.

4.4 Closest prior work

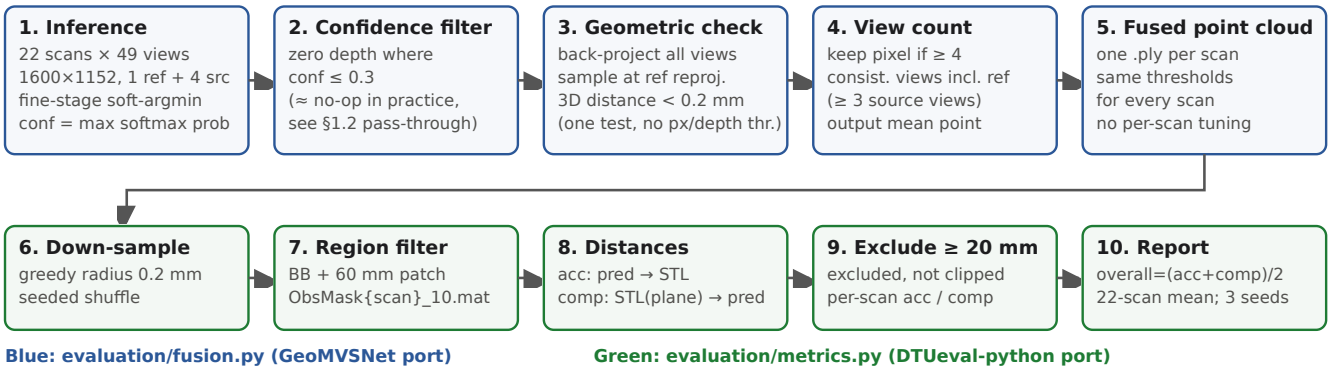
Work	Supervision	How geometry / the monocular prior is used	Relation to this proposal
MonoMVSNet Jiang et al., ICCV 2025, arXiv 2507.11333	Supervised (GT depth)	Reference-view monocular (DA-V2) features fused into the MVS features, with attention to source views via a cross-view position encoding; aligned monocular depth updates depth candidates at edges; relative consistency loss.	Same family as Track 1's feature injection. Track 1 differs only in being self-supervised.
M2Depth Lee et al., arXiv 2608.20788, Aug 2026	Supervised (decayed $\ell 1$ to GT depth; DTU, BlendedMVS)	Frozen Depth Anything V2-Small coupled with cascade MVS by bidirectional mutual refinement; prior-guided cost-volume refinement with attention and discretized depth bins. Reports DTU 0.301 / 0.241 / 0.271 (own protocol).	Supervised. Same frozen DA-V2-S model as Track 1.
DFM-MVS Zhu et al., arXiv 2504.11845, 2025	No real-world labels (pseudo-supervised)	A depth foundation model's prior simulates stereo correspondences for pseudo-supervised training, and guides a correction of coarse-stage errors to limit propagation.	Closest in setting (label-free). Uses the prior as a training label and for correction, not as frozen encoder features in a photometric objective.
GeoMVSNet Zhang et al., CVPR 2023	Supervised	Geometry from *coarse cascade stages* (not an external prior) fused into finer-stage features via a two-branch geometry fusion network; coarse probability volume embedded in regularization; frequency-domain filtering, curriculum; GMM-based depth-distribution loss.	Source of our fusion protocol. Its geometry awareness is internal to the cascade.
CL-MVSNet ICCV 2023, arXiv 2503.08219	Self-supervised	No monocular prior: image- and scene-level contrastive consistency, L0.5 photometric loss.	Self-supervised reference. Its ICC/SCC consistency was ported here: no gain on scan48 (§1.2). Point clouds we generated locally at 640×480 from its released checkpoint are the evaluator-sensitivity case in Track 2(a).

Descriptions are from the papers' abstracts and, for M2Depth and MonoMVSNet, their method sections (checked 2026-10-10). Published DTU numbers use each paper's own fusion and are not comparable to our evaluator (Track 2(a)).

5. The evaluator, documented from the code

Every fused number in this report comes from one in-repo pipeline. The values below were read from the code at `main @ 5572daf`, not from memory.

5.1 Pipeline and invocation



Invoked as: python -m evaluation.eval_dtu --config <cfg> --ckpt <ckpt_final.pt> --out <run>/seedN/eval_1600x1152.json (last step of every queue job)
GT: official STL points, ObsMask, Plane (evaluation/dtu_gt.py). Port cross-checked on scan1 vs the unmodified DTUeval-python/eval.py (2026-09-24).
Deviation: consistent align_corners sampling (flag reference_sampling_quirk reproduces GeoMVSNet exactly).

Figure 11. The evaluator as implemented in this repository (values read from evaluation/fusion.py, evaluation/metrics.py, evaluation/eval_dtu.py at 5572daf). Blue boxes = fusion; green boxes = official metric.

Step	Code	Setting (as in the code)
Invocation	`evaluation/eval_dtu.py`	`python -m evaluation.eval_dtu --config <cfg> --ckpt <run>/seedN/ckpt_final.pt --out <run>/seedN/eval_1600x1152.json`. This is the last step of each queue job (e.g. `runs/queue_b2_seeds_nogeo.log`). The 3-seed aggregate is written to `aggregated_1600x1152.json` as mean ± population std.
Test data and resolution	config block `eval.test_data`	`dtu-test-1200`, `layout: test`, `img_wh: [1600, 1152]` (1600×1200 resized, as in GeoMVSNet's test script). All 22 DTU test scans, every reference view (49 per scan). 1 reference + 4 source views from `pair.txt` (`n_views: 5`). The cost volume is depth-chunked under `no_grad` (exact) to fit in memory.
Depth and confidence	`eval_dtu.py::_predict_scan_views`	Fine-stage soft-argmin depth. Confidence = max softmax probability over the fine-stage hypotheses (GeoMVSNet's `photometric_confidence`).
Fusion	`evaluation/fusion.py::geometric_consistency_fusion`	A port of GeoMVSNet `fusions/dtu/_open3d.py`, run as `scripts/dtu/fusion_dtu.sh` does. For each reference view: zero depths with confidence ≤ `prob_thresh`; back-project every view, the reference included; sample each view's 3D point at the reference pixel's reprojection; count views whose point lies within `dist_thresh` of the reference point; keep pixels with ≥ `num_consist` such views; output the mean of the consistent points.
Fusion thresholds	`GEOMVSNET_PROB_THRESH`, `_DIST_THRESH`, `_NUM_CONSIST`	**Confidence > 0.3. Geometric check: world-space distance < 0.2 mm between the reference point and the reprojected source point. View count ≥ 4 consistent views, the reference itself included (≥ 3 source views).** There is no separate pixel-reprojection threshold and no separate depth-difference threshold: the single 3D-distance test plays both roles. The thresholds are the same for all scans; there is no per-scan tuning. The config can override them (`eval.fusion`), but no reported run does.
Deliberate deviation	`reference_sampling_quirk=False`	The reference normalizes coordinates for `align_corners=True` but samples with `align_corners=False`, a shift of up to 0.5 px. Ours samples consistently. The flag reproduces the reference exactly. At 640 px the shift is about the whole 0.2 mm threshold; at 1600 px it is about 0.05 mm.
Ground truth	`evaluation/dtu_gt.py`	Official DTU `Points/stl`, `ObsMask/ObsMask{scan}_10.mat` (ObsMask, BB, Res) and `Plane{scan}.mat`, from `gt_root` = `../dtu_groundtruth`.
Official metric	`evaluation/metrics.py::dtu_official_metrics`	An in-process Python port of DTUeval-python, the Python version of the MATLAB protocol used by MVSNet, CasMVSNet and GeoMVSNet. No subprocess or MATLAB call. Steps: greedy radius down-sampling at 0.2 mm (seeded shuffle; the official script is unseeded); keep points inside BB + 60 mm patch and inside the ObsMask; accuracy = prediction → STL; completeness = STL above the plane → prediction. Distances ≥ 20 mm are *excluded*, not clipped. Overall = (acc + comp) / 2, averaged over the 22 scans.
Validation of the port	One-off check, recorded in project notes	Cross-checked on scan1 against the unmodified `../DTUeval-python/eval.py` (2026-09-24). It also reproduces the per-scan numbers in `../official_paper_results.txt`.
Validity	—	Numbers from before 2026-09-23 used no fusion filter and a non-official metric. They are void.

5.2 Our fusion vs CL-MVSNet's fusion

Setting	Ours (GeoMVSNet port)	CL-MVSNet `pcd` filter (`../cl_mvsnet_eval/main.py` defaults, `filter/pcd.py`)
Confidence	> 0.3	> 0.8 (`--conf`)
Geometric check	3D distance < 0.2 mm	Pixel reprojection distance < per-scan threshold (0.15–0.75 px, dictionary in `pcd_filter_worker`) *and* relative depth difference < 0.001 (`--depth_thres`)
View count	≥ 4 incl. reference	≥ 3 source views (`--thres_view`)
Per-scan tuning	None	Yes (reprojection threshold)
Test resolution	1600×1152	Script: `--img_size 1184 1600`. The locally generated depth maps are 640×480 (see §5.3).

5.3 CL-MVSNet re-evaluation

> **CL-MVSNet point clouds generated locally at 640×480 from CL-MVSNet's released checkpoint (not released point clouds) score 0.438 under our evaluator vs 0.329 as published. The 0.438 vs 0.329 gap mixes test resolution (640×480 here vs the 1184×1600 set in CL-MVSNet's test script), the evaluation crop, fusion thresholds and checkpoint differences; its cause is unconfirmed.**
> Our evaluator: 0.446 / 0.431 / **0.438** (acc / comp / overall, 22 scans). Published: 0.329. We have not verified the published 0.329.

Provenance of the 0.438 (read from disk for this report). The .ply files in `../cl_mvsnet_eval/outputs/` were generated locally on 2026-06-20 by running CL-MVSNet's released pretrained model and its own `pcd` filter, then scored with the official DTU script (`../eval.sh` → `../official_paper_results.txt`, mean of 22 scans 0.4456 / 0.4312 / 0.4384). Our in-repo metric port reproduces those per-scan numbers. The saved depth maps are 640×480, while CL-MVSNet's test script sets `--img_size 1184 1600`. Test resolution is therefore a second, unexamined candidate cause of the gap, alongside fusion settings. Neither has been isolated (E4).

Consequence for this proposal: every comparison in this report is made under one evaluator, so comparisons between our runs are valid. Absolute numbers are not comparable to published DTU tables. Track 2(a) turns this into a measured finding (E4, E5 in §6.1).

6. Expected outcome and remaining experiments

6.1 Remaining experiments

#	Experiment	Answers	Status
E1	C1_capacity seed 1 (full chain + 1600×1152 eval), then the 3-seed aggregate	Track 2(b): C1 mean ± std vs A0f 0.491 ± 0.009	**PENDING**: running; ~10-11 07:30 UTC
E2	B2_mono_feat seed 2 vs B2_nogeo seed 2	Track 1 screen	**PENDING**: starts ~10-11 07:30 UTC, ~4 h
E3	B2_nogeo seeds 0 and 1	Track 2(c); whether (d) needs scoping (§4.1 box)	**PENDING**: not queued; ~1 day each
E4	Score A0f and C1 depth maps under CL-MVSNet's fusion (per-scan thresholds), and isolate the cause of the 0.438 vs 0.329 gap: re-run CL-MVSNet at its scripted 1184×1600 and swap fusions on fixed depth maps	Track 2(a), "report both evaluators"	**PENDING**: not started; no training
E5	C1 scored at the 640×512 crop vs full frame	Track 2(a), crop effect on a second model	**PENDING**: not started; no training
E6	Why C1 fuses better while its per-view depth is worse	Track 2(b) mechanism	**PENDING**: open
E7	B2_mono_feat seeds 0 and 1	Track 1 claim	Only if E2 shows a gain > 0.009
E8	GeoCorr Lite v0 vs its α = 0 control	Whether a score-level geometry correction helps	**NOT RUN**: proposed next experiment

6.2 Expected outcome

- **Track 2** delivers a paper whose contributions are measured findings: evaluator sensitivity (a), capacity as the limit (b), a component analysis with seed noise (c), and a scoped negative result for the monocular-anchored prior (d). Most of the evidence already exists. The missing pieces are E1, E3, E4 and E5. Final C1 3-seed numbers, the both-evaluators table and the L_geo seed check are all **PENDING**.
- **Track 1** either adds one narrow self-supervised finding ("frozen DA-V2-S features help without GT supervision") or is reported as a null result. Its outcome is **PENDING** and the claim is not yet established.
- **GeoCorr Lite** enters the paper only if E8 is run and beats its own α = 0 control beyond seed noise. Until then it is **NOT RUN**.
- No expected numbers are given for pending runs. They will be filled in from measurements only.

6.3 Approval requested

> 1. Change the primary contribution from "geometry prior as a network input (GGF + GeoCorr Lite)" to the Track 2 empirical study.
> 2. Run Track 1 as a screen only, with the seed-noise criterion above.
> 3. Queue E3 (B2_nogeo seeds 0, 1) and E4/E5 (evaluator work, no training) after the current queues.
> 4. Decide whether E8 (GeoCorr Lite v0 vs α = 0) runs after E1–E3.

Sources (internal): runs/A0f_fixes_only_short/{aggregated_1600x1152.json, seed*/eval_1600x1152.json}; runs/A0_baseline/{aggregated.json, aggregated_1600x1152.json}; runs/B2_capacity/seed{0,2}/{eval_1600x1152.json, ckpt_final_costvol_diag.json, probe_final/, val_depth.jsonl}; runs/B2_nogeo/seed2/ (same files); runs/prior_vs_dips/, runs/prior_quality/, runs/prior_confidence_val/, runs/prior_confidence_val_B2/, runs/fusion_pass_diag/; runs/queue_b2_seeds_nogeo.log; NOTES.md; experiment-history doc. ../official_paper_results.txt (CL-MVSNet plys under our evaluator).

External: Jiang et al., MonoMVSNet, ICCV 2025, arXiv 2507.11333 · Lee et al., M2Depth, arXiv 2608.20788 · Zhu et al., DFM-MVS, arXiv 2504.11845 · Zhang et al., GeoMVSNet, CVPR 2023 · CL-MVSNet, ICCV 2023, arXiv 2503.08219.